Digital Clock Project

Progress Log

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1 Project Start

We started the Digital Clock project by understanding the requirements and discussing how the system could be implemented. Both members contributed to the initial planning and discussion.

G.Rithvik

- Worked on the programming part that is required for the clock.
- Helped in construction the circuit.
- Contributed to the initial circuit planning.

R.Srikar

- Helped understand the project requirements.
- Researched about the apparatus and connections of circuit.
- Contributed to documenting the whole project.

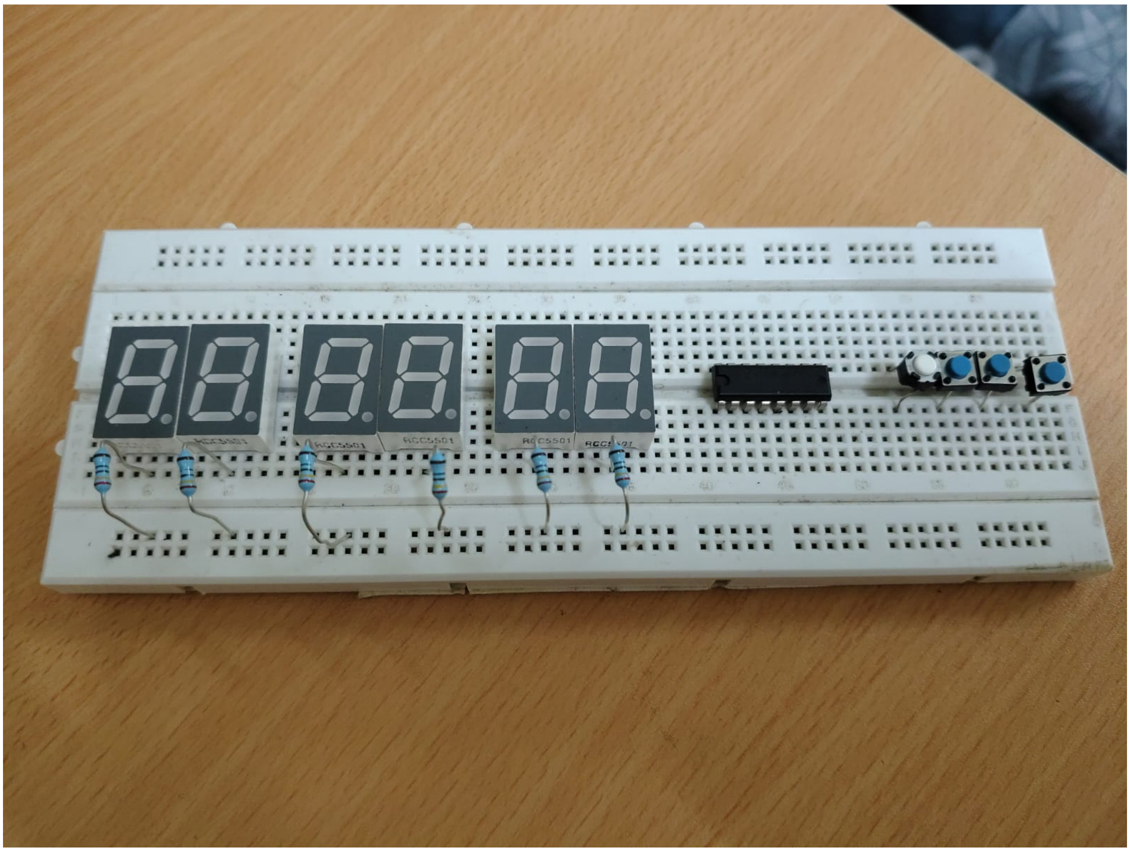

Figure 1: Initial project setup and planning

2 Initial Circuit Setup

We began assembling the Digital Clock circuit according to our planned design. We checked the connections and tested the circuit step by step.

G.Rithvik

- Worked on assembling the circuit.
- Checked the connections between the components.
- Helped test the initial circuit.

R.Srikar

- Assisted with circuit assembly.
- Checked the power and component connections.
- Helped identify and correct connection issues.

Problem faced: We got a problem with the displays.They did not show the digits properly

Solution: It was the connection issue.we reconnected the jumpers again.

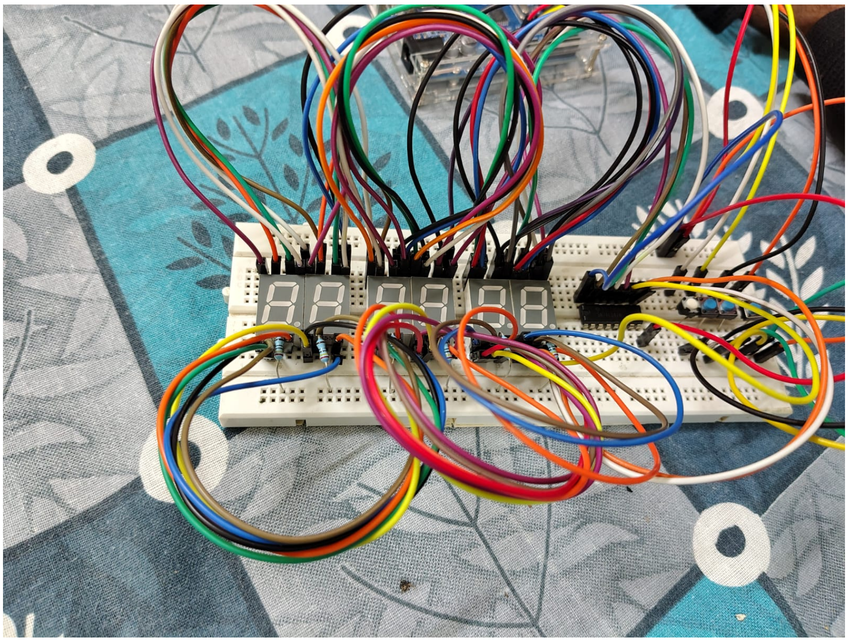

Figure 2: Initil circuit assembly

3 Programming and Testing

After assembling the circuit, we worked on the Arduino program and tested it with the hardware.

G.Rithvik

- Worked on the Arduino code.
- Tested the program with the circuit.
- Helped debug errors in the program.

R.Srikar

- Assisted in testing the program.
- Checked whether the displayed time was correct.
- Helped identify problems during testing.

What we learned:
K-Maps, State Machines, Multiplexing, Circuit Building, Code

4 Debugging

During testing, we encountered some problems with the circuit. We worked together to identify the causes and make the required changes.

Problem

The problem we encountered was the displays were not displaying the digits properly.There were also some sections that wouldn't get power/light up.

Our Approach

Our approach to our problem we encountered is we unpluging all the jumper wires and checking whether all the displays are working with a working jumper wire.There were some broken jumper wires we removed them.

Solution

We connected the jumper wires in a proper manner to the breadboard.we figured out that the issue was loose connection of jumper wires and some faulty jumper wires

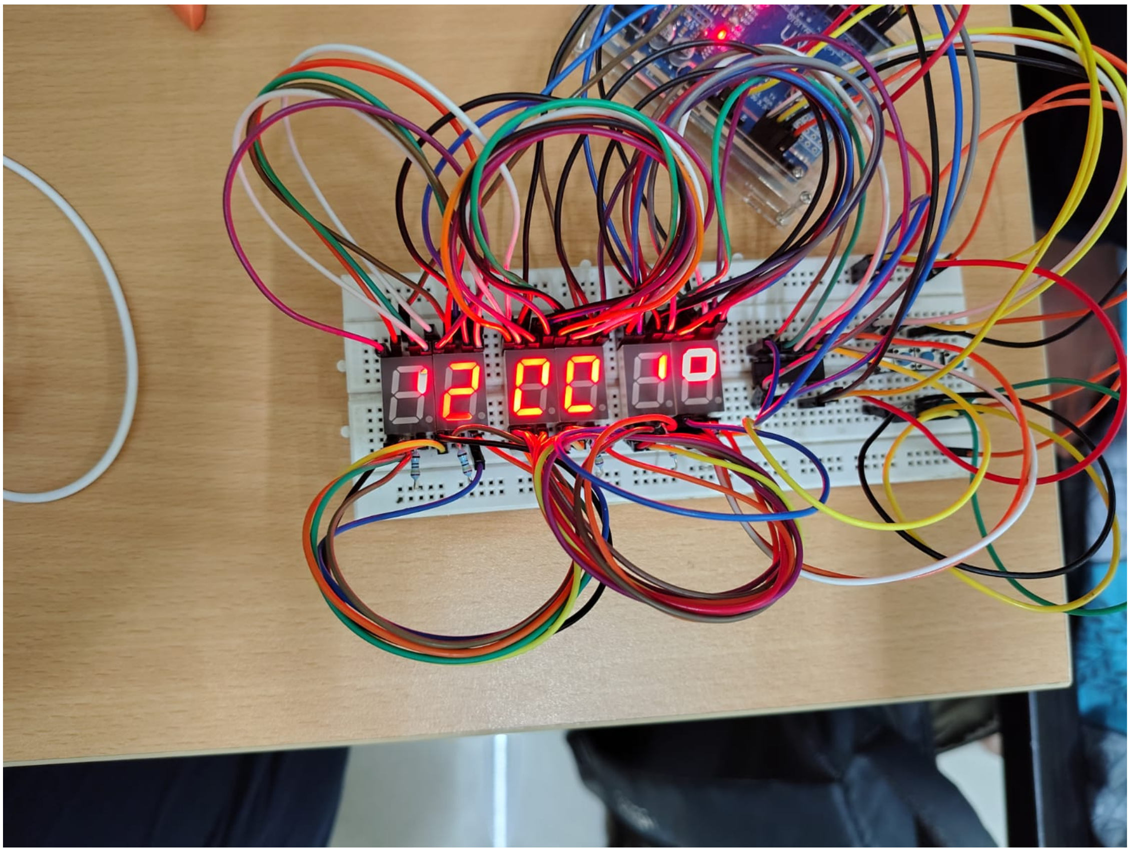

Figure 3: Testing the Digital Clock

5 Final Testing

After making the required changes, we tested the complete Digital Clock. Both members checked the operation of the system and verified the output.

G.Rithvik

verified the arduino code.

R.Srikar

Checked whether the displayed time was correct.

Labelling of the push buttons:

- from left to right the push buttons are labelled as 0,1,2,3

Function of push buttons:

- 0 → Start/Pause
- 1 → Next

- 2 → Increment
- 3 → Decrement

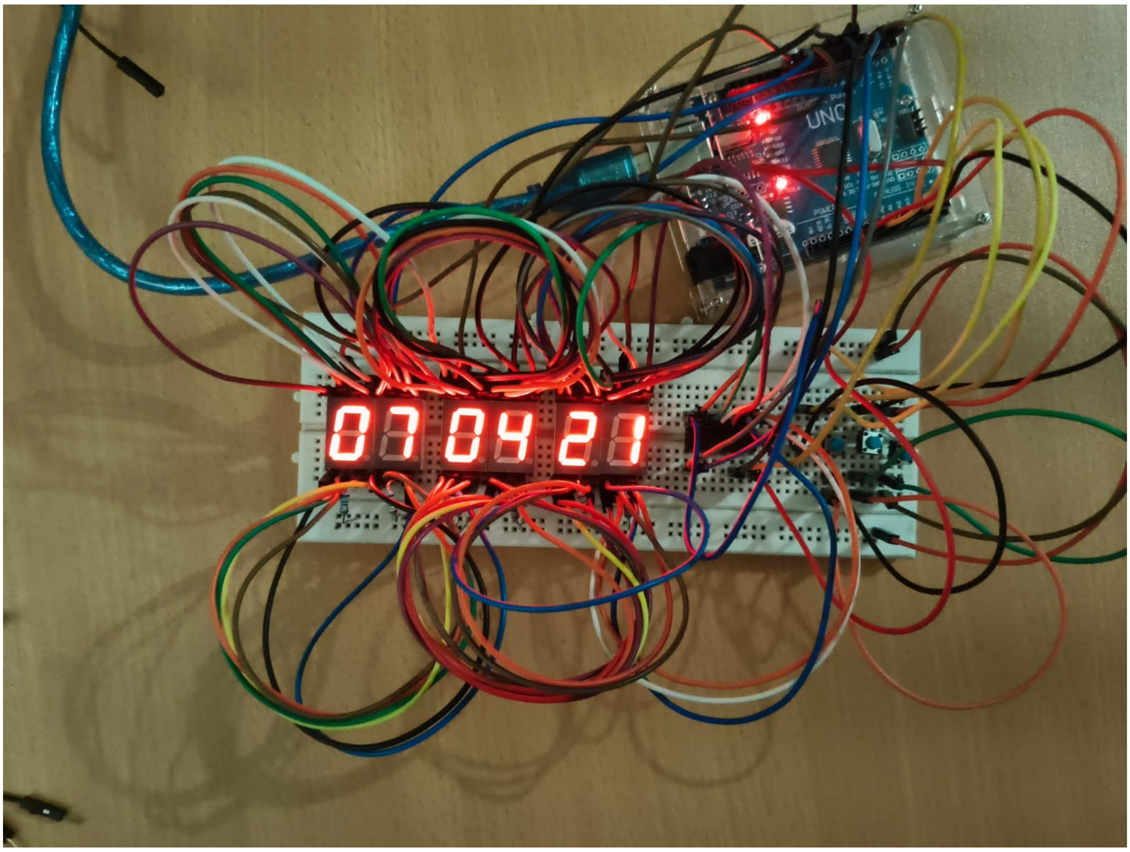

Figure 4: Final working Digital Clock

6 Individual Learning

G.Rithvik

Through this project, I learned about the implementation and working of a Digital Clock. I also learned about **K-Maps and its uses, circuit assembling, uses of components used in the making of digital clock.**

The most difficult part for me was **Writing and understanding the code which is used in the hardware**

The most interesting part was **Figuring out the connections made between the components in the circuit**

R.Srikar

Through this project, I learned about the implementation and working of a Digital Clock. I also learned about **K-Maps and its uses, circuit assembling, uses of components**

used in the making of digital clock.

The most difficult part for me was **Figuring out the mistakes made in connecting the components**

The most interesting part was **Figuring out the connections made between the components in the circuit**

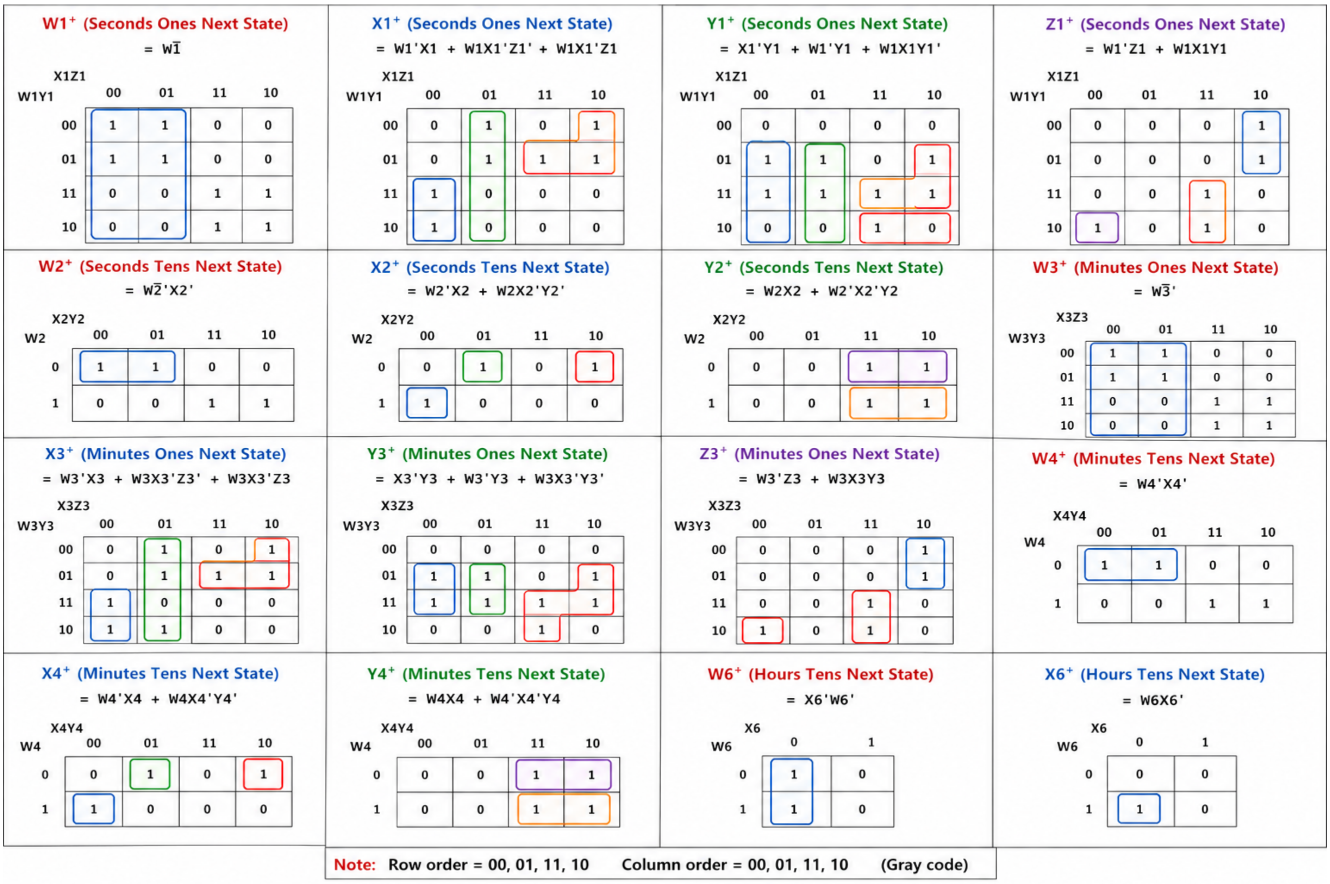

Figure 5: K-Maps of the code

7 Final Reflection

Working as a two-member team allowed us to divide the work and solve problems together. The project involved planning, assembling the circuit, programming, testing, debugging, and final verification.